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Small molecule conjugation reduces macrophage uptake and increases *in vivo* blood circulation of polystyrene nanoparticles

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Supplementary material for this article is available [online](#)

Abstract

Nanomedicine often failed clinically to show therapeutic efficacy due to reduced particle circulation and enhanced capture by the reticuloendothelial system, including the liver. Developing novel immunomodulatory surface coating can prevent macrophage capture and increase the particle circulation of the nanomedicine, resulting in higher therapeutic efficiency. Herein, we demonstrate the development of immunomodulatory small molecule (RZA15) with triazole functionality using copper-catalyzed click chemistry to conjugate onto spherical polystyrene nanoparticles (NPs) using amide coupling reactions, achieving higher blood circulation and lesser macrophage uptake of the nanoconjugates (NCs). In this work, we evaluated the effectiveness of RZA15 coating for the enhanced circulation of polystyrene NPs of 100 nm size, which is commonly utilized for various drug delivery applications, and compared with poly(ethylene)glycol coatings. Several polystyrene NC formulations were analyzed *in vitro* in normal and macrophage cells for cell viability and cellular uptake studies. *In vitro* studies demonstrated lesser macrophage uptake of the NCs following RZA15 coating. Finally, *in vivo*, blood-circulation, pharmacokinetics, and biodistribution studies were performed in the C57BL/6J mouse model that endorsed the substantial role of RZA15 in reducing liver and spleen capture and results in extended circulation in blood. Coating immunomodulatory small molecules to nanoparticles can severely enhance the potential therapeutic effects of nanomedicine at lower doses in near future.

1. Introduction

Delivery of therapeutic agents to the tumor site is critical to avoid drug toxicity associated with free circulating drugs in the body and lack of targeting. This has further led to drug resistance and introduced other side effects [1]. Nanoparticle (NP)-mediated drug delivery has vast possibilities because of their target specificity, better circulation, and retention, better bioavailability, slow and sustained release of therapeutic agents, lower toxicity and enhanced therapeutic efficiency, etc [2, 3]. However, nanomedicine often fails to show *in vivo* therapeutic efficacy at low doses due to lack of targeting, reduced particle circulation, particle retention, and enhanced capture by the reticuloendothelial system (RES), including liver and spleen. Active targeting agents have been a

popular approach to increase NPs uptake in the target disease site. Even though the active targeting strategy using targeting agents improved the biodistribution of NPs in blood and target sites, less than 1% of administered NPs can only reach the target tumor site [4, 5]. This has generated a significant challenge for the researchers to deliver low therapeutic doses, avoiding long-term toxicity, drug resistance, immunogenicity, etc. Various strategies are currently being explored, including pegylation techniques. Pegylation of NPs using various molecular weights (MWs) of polyethylene glycol (PEG) has become a common strategy to elevate the circulation of NPs, reduce opsonins binding, and fast clearance through excretory systems [6].

Host immune response plays a vital role in the recognition of foreign material when introduced

to the human body, further controlling circulation and biodistribution of any injected material [7]. Macrophage or phagocytic capture reduces particle circulation in blood. Developing immune-invasive coating can prevent immune recognition and reduce macrophage capture, enhancing the NPs circulation in the blood [8, 9]. Vegas *et al* discovered a triazole containing a small molecule (RZA15) through a combinatorial screening approach that [10] leads to reduction of the recruitment and attachment of the macrophages [11]. This was confirmed using intravital imaging *in vivo*, was instrumental in establishing the hypothesis that this novel class of small molecules can be used as an immunomodulatory agent to coat the NPs, helping in reduced immune recognition, macrophage uptake, and improved blood circulation and targeting to diseased site [12, 13].

In this context, we evaluated the effectiveness of an immunomodulatory small molecule (RZA15) coating for the enhanced circulation of polystyrene nanoparticles with 100 nm size, which is usually applied for numerous drug delivery applications. Further, we compared the efficacy of this coating with PEG coatings as a structural control, which is currently used by many research groups and in approved formulations. Detailed *in vitro* studies were performed demonstrating the reduced uptake of RZA15 coated polystyrene NPs (PR) formulations in macrophage cells than uncoated (P) and pegylated polystyrene (PP) formulations [14]. Lastly, *in vivo* blood-circulation, pharmacokinetics (PK) and biodistribution studies were performed in the C57BL/6J mouse model, showing less liver and spleen capture in the RZA15 coated group and resulting in better circulation. Coating immunomodulatory small molecules to NPs can severely enhance the potential therapeutic effects of nanomedicine at lower doses [15].

2. Experimental procedures

2.1. Materials

COOH functionalized polystyrene NPs were purchased from Phosphorex, Inc. PEG400, 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride (DMTMM) reagents were purchased from Sigma. Dulbecco's modified Eagle's medium (DMEM), ethylene diamine tetra acetic acid (EDTA), trypsin, dimethyl sulfoxide (DMSO), and 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) dye were purchased from Thermo Fisher Scientific. Murine macrophage RAW 264.7 and human ARPE-19 cells were obtained commercially from the American type culture collection.

2.2. Synthesis of RZA15

RZA15 (Z1-Y15) was synthesized according to the previously described protocol [11, 16]. A round bottom flask (250 ml) filled with methanol and water

(5:1) was added with 4-propargylthiomorpholine 1,1-dioxide (1 eq.), then 25 equivalents of tris[(1-benzyl-1H-1,2,3-triazol-4-yl)methyl]amine, copper iodide and triethylamine were mixed. After purging this mixture for 15 min with argon, it was brought down to 0 °C and further added with 11-azido-3,6,9-trioxaundecan-1-amine (1 eq., 6.30 g, 28.86 mmol). At room temperature, the mixture was stirred for 15 min, then heated to 55 °C overnight. It was cooled at room temperature and filtered by Celite to get rid of insoluble material. With the help of rotavap using silica under optimized pressure, the filtrate was dried. Using 120 g ISCO silica column liquid chromatography by dichloromethane: ultra (22% MeOH in dichloromethane (DCM) with 3% NH₄OH) mixture 0% to 40% for purification and then characterized by nuclear magnetic resonance (NMR) mass spectroscopy and electrospray ionization mass spectrometry (ESI-MS) [12, 17]. NMR was performed using Bruker AVANCE III HD 600 MHz. ESI-MS was performed using MS: Bruker Micro ToF ESI LC-MS System. X-ray photoelectron spectroscopy (XPS)-spectrometer was performed using PHI Quantera XPS. Dynamic light scattering (DLS) by Zetasizer Ver. 6.20, Malvern Instruments Ltd [18].

2.3. Conjugation of PEG and RZA15

The 100 µl polystyrene NPs with COOH functionalization (as purchased from the vendor) were taken and diluted into 1 ml Mili Q water. Following, 10 mg ml⁻¹ PEG400/RZA15 was added dropwise to the reaction mixture. After 5 min of stirring, DMTMM solution (5 mg ml⁻¹) was added dropwise under constant stirring. The reaction mixture was stirred for 4 h. Covalent ester bonds were responsible for the PEG coating to the NPs, whereas the covalent amide bond for RZA15 coating to the NPs. Before, they were dialyzed against Mili-Q water using dialysis cassettes for 1 h. The purified nanoconjugates (NCs) were collected and used for all characterization and biological studies.

2.4. Cell viability assay

Colorimetric MTT assay is an approach for assessing the cytotoxicity of any NPs or compound using cultivated cells. MTT dye with the action of mitochondrial dehydrogenase enzymes reduces from yellow to purple formazan crystals [19, 20]. Mouse macrophage (RAW 264.7) cell line cultured in 96 well plates (10 000 cells per well) incubated for 24 h. These cultured cells were treated with NPs coated with small molecule RZA15 (PR) at different doses (5e⁶–5e¹¹ polystyrene particles) and further incubated for 24 h. By replacing the previous media, MTT reagent in DMEM (0.05 mg well⁻¹) was added to each well. DMSO and methanol in a 1:1 ratio were used to solubilize the formazan crystals. The purple color was measured in a microplate reader at 570 nm absorbance. The results were expressed as

percent cell viability = $([A570(\text{treated cells}) - \text{background}] / [A570(\text{untreated cells}) - \text{background}]) \times 100$ [20].

2.5. Cellular uptake studies

Uptake ability was examined using RAW 264.7 and ARPE-19 cells (40 000 cells/24 well plate) in 500 μl media. These cells were incubated with NPs (P), pegylated NPs (PP), and RZA15 NPs (PR) at a dose of $5 \times 10^{11} \text{ ml}^{-1}$. Three replicates were kept for each group ($N = 3$). At different time points (20 min, 45 min, and 4 h), cells were washed and lysed using lysis buffer, and fluorescence intensity corresponding to the NPs was measured using a microplate reader (Tecan). Based on the standard curve of each particle, the extent of particles uptake per cell was calculated and plotted. Further incubating 4 h, the medium was replaced and again incubated for 24 h. Fluorescence microscopy was performed to detect the uptake of cells and NC [21–23].

2.6. *In vivo* blood distribution studies

All *in vivo* experiments were performed following Institutional Animals Ethics Committee (IAEC) approval. Bio-compatibility assessment of materials and device in model has been approved by IAEC of Indian Institute of Technology, BHU, Varanasi approval ID: IIT(BHU)/IAEC/2023/067. The NPs 10^{11} ml^{-1} were injected intravenously through the tail vein using a restrainer chamber in the C57BL6 mice and bled through a saphenous vein at scheduled intervals. To prevent blood clotting, anticoagulant tubes containing sodium heparin were used. Using cardiac puncture, terminal blood samples were obtained. Red blood cell lysis buffer (Thermo) was used for blood dilution 1:20. After filtration of the lysate with the filter flow cap tubes, a flow cytometer (Attune NxT) was used to examine the Cy5 channel signals. Sequential processing was done from (a–d) using BD FACSCanto™ Software. C57BL6 mice (CRL) blood treated with heparin was used to determine the background and spikes of Cy5 bead concentration. All organs were isolated for the termination of the experimental groups and processed through an *in vivo* bioimager (Perkin Elmer). Biodistribution studies from the isolated organs were then done by estimating the fluorescence intensity by Fiji (ImageJ) software and further statistically analyzed using ANOVA.

2.7. PK parameters

The PK parameters were analyzed in C57BL/6J mice administered with NPs P, PP, and PR (10^{11} ml^{-1}). The IV infusion estimated PK parameters, non-compartmental PK model in PK solver.

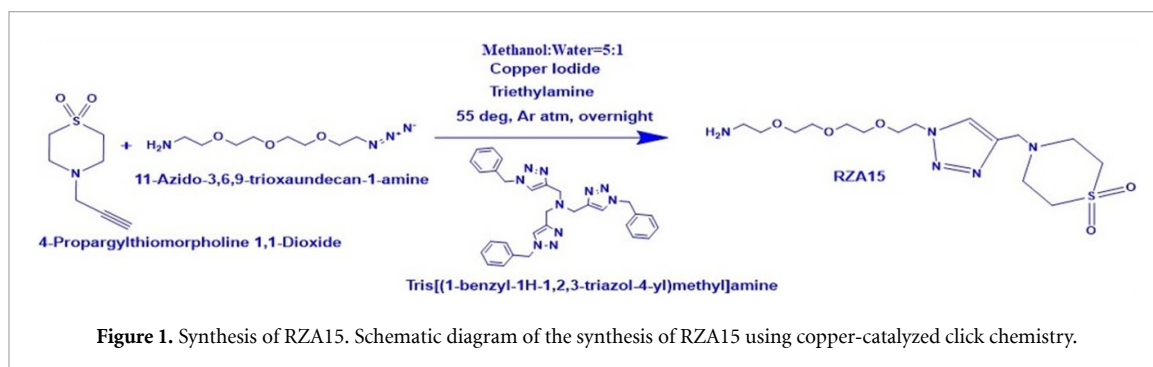
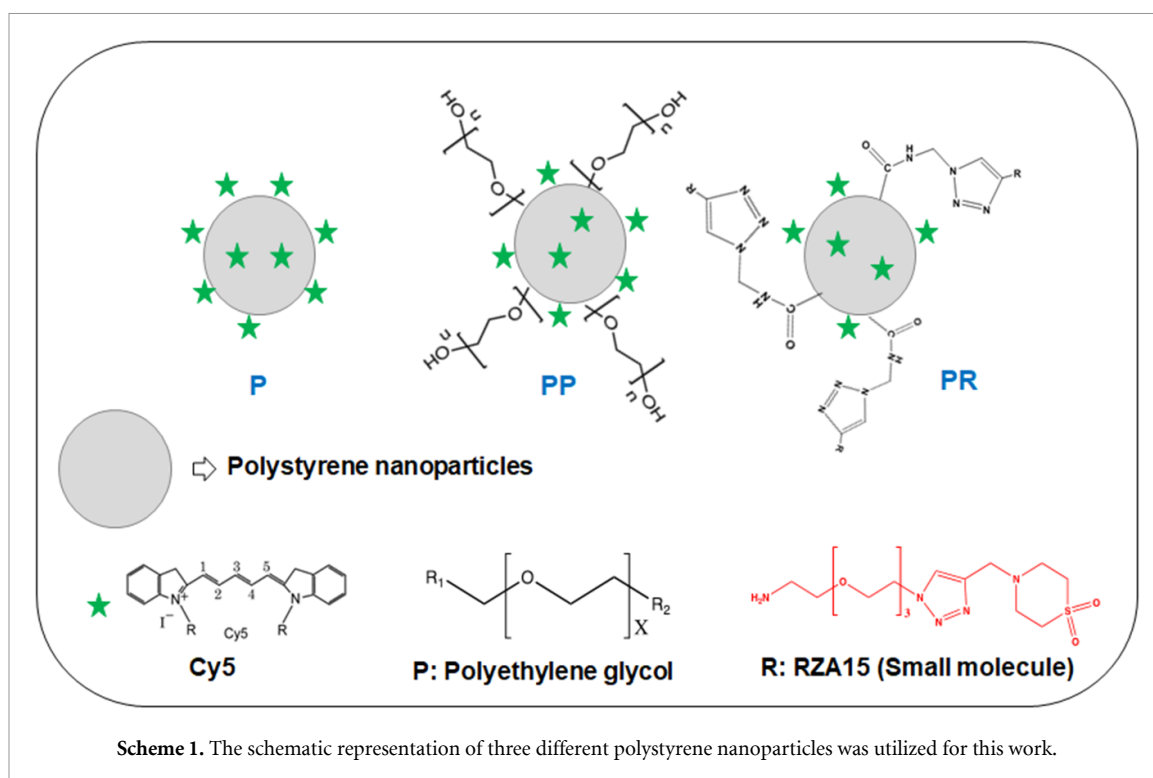
3. Results

Macrophage recognition, attachment followed by RES captures often reduce their blood circulation, affecting their lack of targeting and efficacy [24]. In previous studies, RZA15 showed immune-protective properties reducing macrophage recognition and attachment when coated with hydrogel and medical implants [11]. Mechanistic confirmation demonstrated the lesser recruitment of macrophages following the coating of RZA15 to hydrogels, which has encouraged us to use RZA15 to coat near-IR polystyrene NPs (purchased from Phosphorex, Inc.) with 100 nm size. The hypothesis of this study is to show that RZA15 coating to NPs may reduce the macrophage recognition and consequent capture of these NPs, improving the blood circulation and therapeutic efficiency of the potential drug delivery systems (DDSs).

First, the small molecule (RZA15) was synthesized as per published methods using copper-catalyzed click chemistry reaction (scheme 1, figure 1(a)) [25]. The synthesized small molecule (RZA15) was thoroughly characterized using NMR and mass spectroscopy ($M + H^+ = 392.21$), confirming their formation and purity (figures 1(b)–(d)).

COOH-functionalized polystyrene NPs (PR) were conjugated with amine-containing RZA15 using an amide coupling reaction using DMTMM as a coupling agent. The NCs were then purified using extensive dialysis before *in vitro* and *in vivo* analysis to avoid any unattached small molecule residues. Unmodified polystyrene NPs (P) were directly purchased from Phosphorex, Inc. and further dialyzed on the same time frame to maintain equal fluorescence intensities tested using a microplate reader.

PEG-400 with MW of 400 was attached to the COOH-functionalized polystyrene NPs using similar chemistry, leading to PP and undergoing dialysis for purification using similar procedures. PEG-400 was used as a structural control to have similar molecular weight like RZA15 that has smaller PEG chain in their backbones. All the NPs and NCs were Cy5 tagged, enabling the formulation to be fluorescent active and useful for nanoparticle tracking to analyze the uptake and distribution *in vitro* and *in vivo*. XPS was performed to confirm the attachment of RZA15 with polystyrene NPs. PR showed a higher amount of nitrogen (N) content compared to the unmodified NPs (P), confirming the attachment of nitrogen containing RZA15 to the polystyrene surface using covalent amide bonding (figure 2). DLS studies confirmed the hydrodynamic diameter and surface charge of the NPs and NCs (SI-table 1). Results demonstrate a minimum increase in the size of NPs following small molecule conjugation (P: $231 \pm 14.9 \text{ nm}$, PP: $241 \pm 8.01 \text{ nm}$ and PR: 243.3 ± 33.8). This indicates



the stability of the NPs even after the conjugation with PEG or RZA15. The stability of nanoparticles in serum for 48 h using DLS was tabulated (SI table 2 and SI table 3). Results show that the size of the NPs incubated in serum has increased from the original size (P: 459.3 ± 16.8 nm, PP: 399.1 ± 19.5 nm, and PR: 409.2 ± 52.6). It was nearly consistent across groups, which indicated reasonable colloidal stability. However, the surface charge has slightly reduced following RZA15 coating, reflecting the conjugation of the $-NH_2$ group (present in the positively charged RZA15 molecule) to the polystyrene NPs. RZA15 coating to polystyrene NPs and administering it in serum has not changed the zeta potential significantly than uncoated polystyrene NPs indicating fare stability in biological media, useful in biomedical applications.

Cell viability was tested using MTT assay in a mouse macrophage (RAW 264.7) cell line for 48 h to assess the effect of RZA15 coating in the *in vitro*

cellular toxicity in a variety of doses ($5e^6$ – $5e^{11}$ polystyrene particles) (figure 3). The cell viability results confirmed that RZA15 coating to polystyrene NPs (PR) did not affect the cell viability of macrophage cells compared to only polystyrene and PEGylated polystyrene groups (P and PP). However, it is important to mention that the polystyrene NPs at this range ($5e^6$ – $5e^{11}$ particles) are cytotoxic to RAW 264.7 cell line (40%–60% cell viability). Hence, optimizing the dose to avoid toxicity will be necessary for future drug delivery applications.

To measure the cellular uptake of polystyrene NPs coated with RZA15, PEG, or uncoated, RAW 264.7 and ARPE-19 cell lines are incubated with various treatment groups (P, PP, and PR). Results demonstrated that RZA15 coating (PR) has significantly reduced the uptake of the nanoparticles in macrophage cells at 20 min, 45 min and 4 h' time after treatment, compared to the human retinal cell line (figures 4(a) and (b), SI figure 2). This

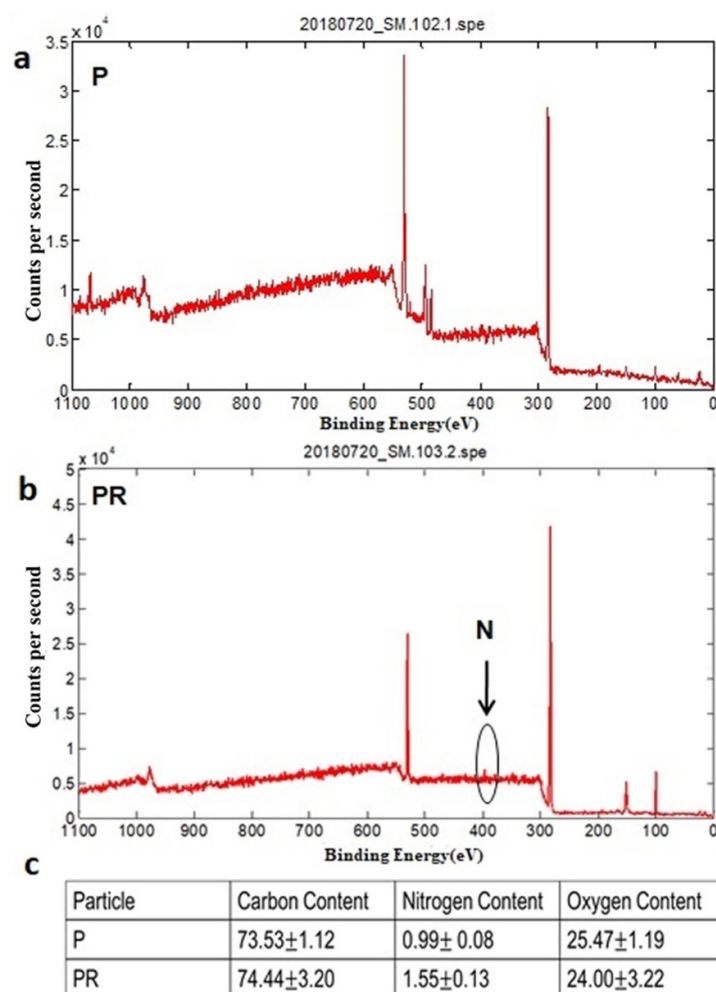


Figure 2. XPS data for (a) P and (b) PR confirming the attachment of RZA15 to the polystyrene nanoparticles; (c) elemental quantification of P and PR from XPS confirming RZA15 attachment.

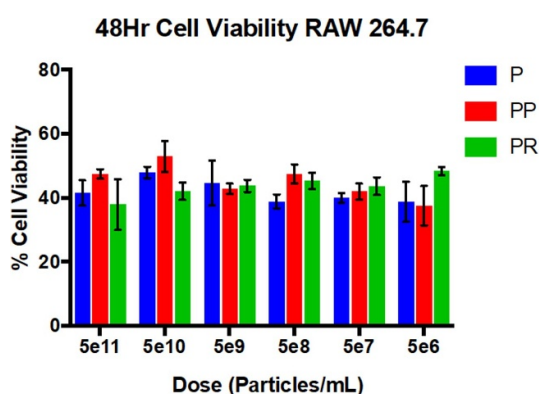
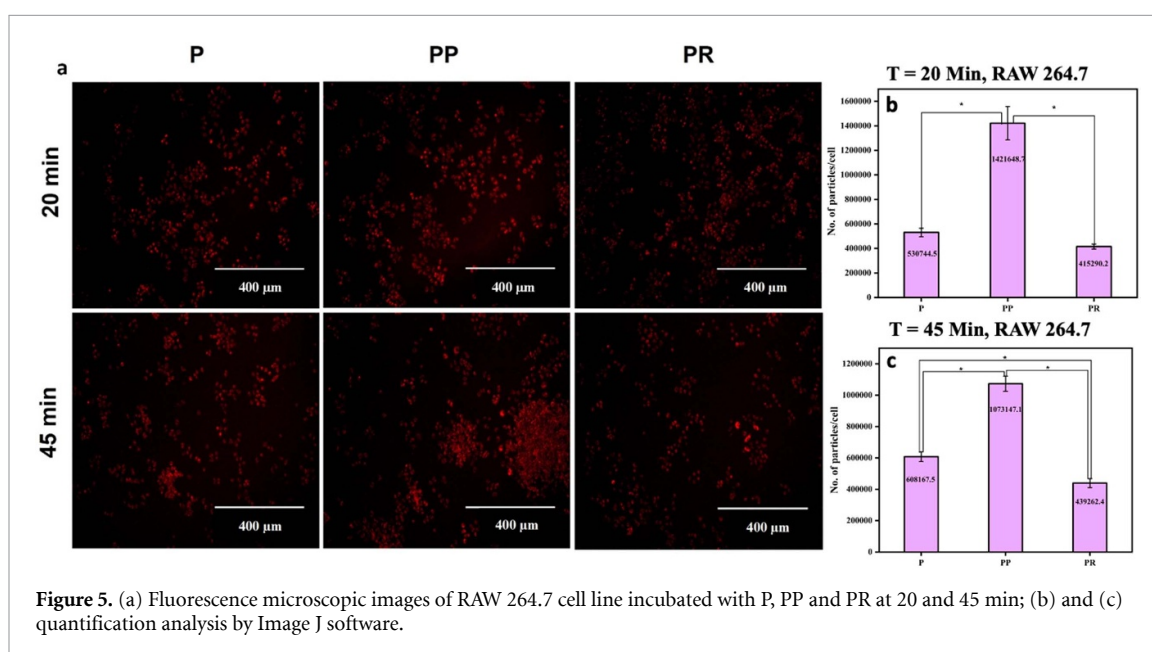
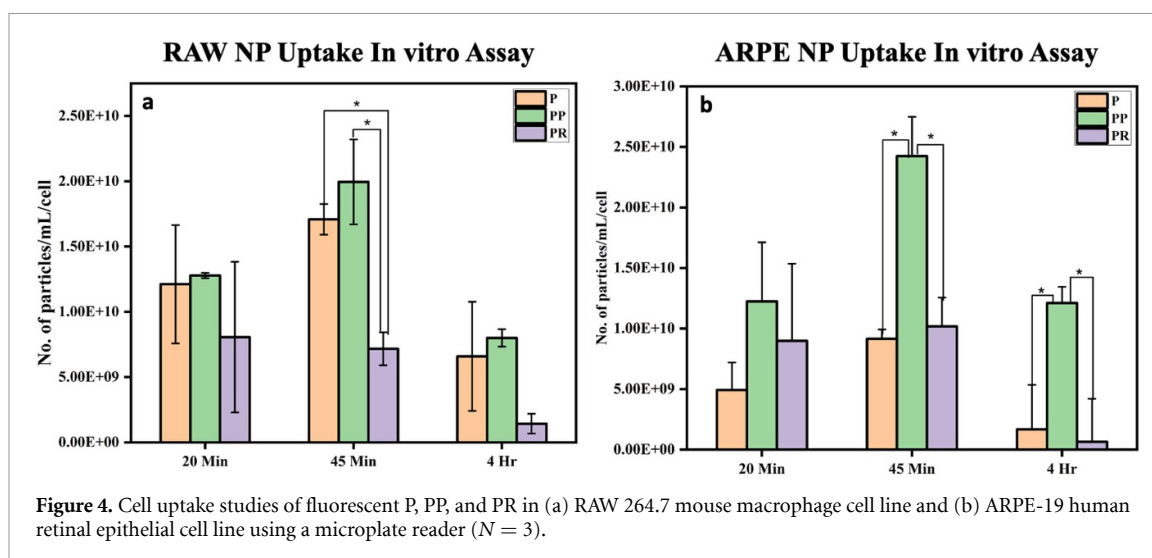


Figure 3. Cell viability studies of P, PP, and PR in RAW 264.7 mouse macrophage cell lines using MTT assay at 48 h in a dose-dependent manner.

is crucial for reducing nonspecific macrophage capture of RZA15 modified nanoparticles that may help to improve blood circulation and enhance targeting and therapeutic efficacy. Fluorescence microscopic analysis was further performed to validate the effects

of RZA15 coating in the reduction of macrophage uptake of PR nanoparticles compared to uncoated polystyrene NPs (P) and PEGylated NPs (PP) at 20 and 45 min (figures 5(a)–(c), SI figure 3). The fluorescence microscopy images, along with the quantification using ImageJ analysis, clearly support the reduction of nanoparticle uptake following coating of RZA15 at 20 and 45 min compared to other treatment groups (uncoated polystyrene NPs: P and PP NPs: PP). These results support the effects of RZA15 small molecule preventing macrophage capture that may be important for enhanced targeting and therapeutic efficiency.

Following confirmation of the *in vitro* efficacy of RZA15 coating in assisting the reduction of NPs' uptake to macrophage cell line, *in vivo* distribution studies were performed in blood following the intravenous treatment using similar groups (P, PP, and PR) to ensure the utility of the RZA15 enhancing the blood circulation and reduction of liver uptake using flow cytometer (Attune NxT) (figures 6(a)–(e), SI scheme 1, and SI. Figure 4). Figures 6(a)–(d) demonstrates the representative 20-minute data for all the



groups showing the detected particle concentration using fluorescence-based FACS analysis. The data confirmed that at 20 min, PR has a more significant number of NPs per ml of blood compared to the other treatment groups (P and PP).

Results demonstrated enhanced blood distribution of RZA15-coated polystyrene nanoparticles in each time point compared to the other treatment groups (P and PP), confirming the improved circulation (figure 6(e)). It is important to note that the average distribution of PR is higher in all the time points compared to P and PP groups even though only 10 and 20 min data point demonstrate statistical significance (figure 6(e)). To further check the biodistribution of the NPs groups in various organs, in vivo imaging system (IVIS) imaging was performed (figures 7(a) and (b)). Results demonstrated reduction in the nanoparticle biodistribution to the organs

(liver, spleen, and kidney) following RZA15 coating compared to uncoated nanoparticles and PEG modified group. A recent paper by Sender *et al* demonstrated that the liver and spleen have a significant higher number of macrophage populations compared to other organs [26]. Our results corroborated accordingly to observed lower uptake in the liver and spleen following small molecule conjugation to the NPs. Hence, lower accumulation of NPs following small molecule coating can be attributed to lower macrophage recognition and capture compared to uncoated NPs.

PK parameters were analyzed through a non-compartmental PK model (table 1). The results show that the half-life ($t_{1/2}$) of the pegylated NPs and RZA15-coated NPs are higher than the uncoated NPs. This signifies that PEG and RZA15 coating has increased the time of circulation of the NPs

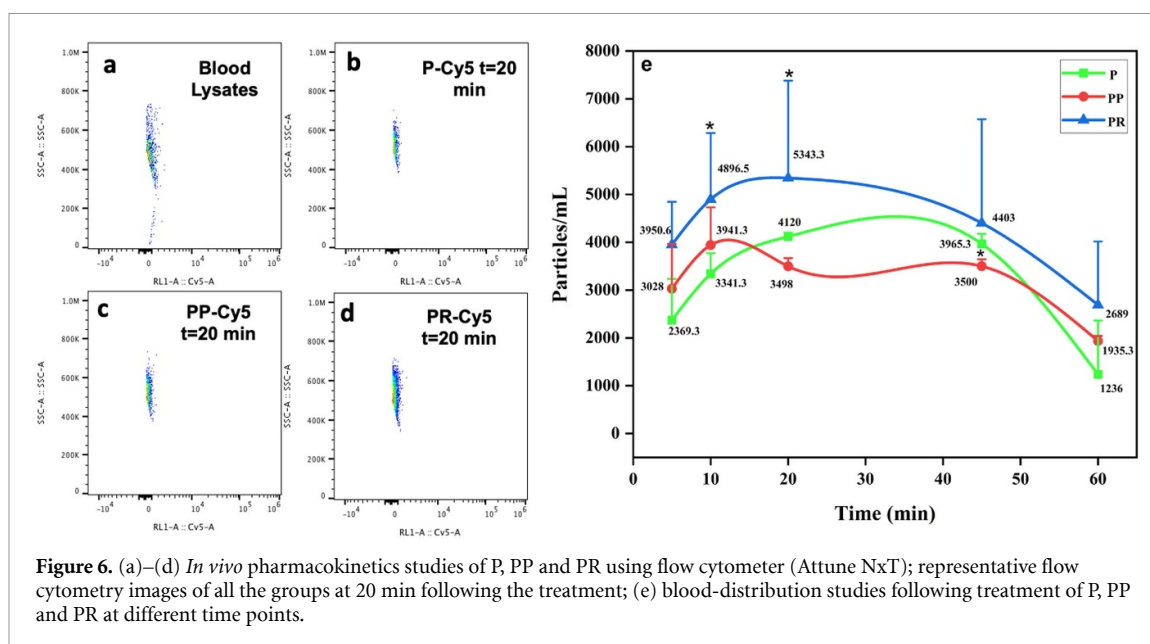


Figure 6. (a)–(d) *In vivo* pharmacokinetics studies of P, PP and PR using flow cytometer (Attune NxT); representative flow cytometry images of all the groups at 20 min following the treatment; (e) blood-distribution studies following treatment of P, PP and PR at different time points.

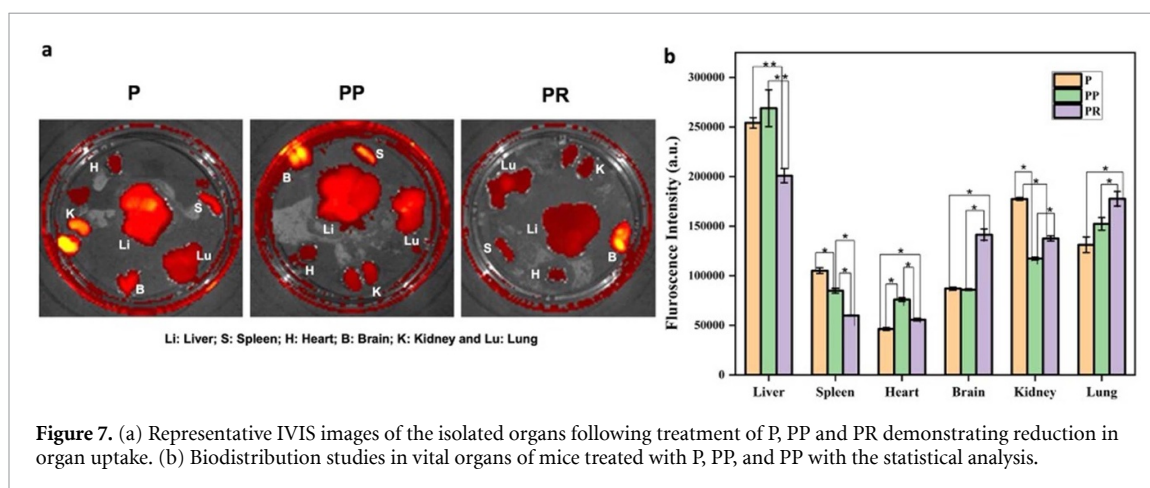


Figure 7. (a) Representative IVIS images of the isolated organs following treatment of P, PP and PR demonstrating reduction in organ uptake. (b) Biodistribution studies in vital organs of mice treated with P, PP, and PP with the statistical analysis.

and reduced the NPs elimination from the plasma compared to the uncoated control. This further indicated a slow and sustained elimination pattern of PP and PR from the blood than uncoated P. Maximum time required to reach the highest concentration (T_{max}) is similar for P and PR. This signifies no significant alteration to the nanoparticles even after the coating of RZA15. Maximum concentration (C_{max}) describes delivery to its maximum concentration at a site for the effective therapeutic response [27]. Here, PR shows the highest C_{max} compared to the other groups, supporting an immediate and effective action compared to P and PP. Higher mean residence time (MRT) represents the average amount of time of NPs. PP and PR show higher MRT than P, which shows NPs coated with PEG and RZA15 help prolong the release or stay in the body for an extended period. Higher MRT also supports a reduction in the rate of NP's clearance, extending the therapeutic window [28]. Moreover, MRT and $t_{1/2}$ correlate with the plasma clearance rate (Cl_{obs}), confirming the lower clearance in the PEG and RZA15

coated NPs groups (PP and PR) compared to the uncoated NPs. PK parameters indicated that the PR-treated group could attain maximum plasma levels in considerable time and show slow and sustained elimination.

4. Discussion

We have reported injectable NPs surface coated with novel immunomodulatory small molecule RZA15 to reduce macrophage capture and enhance circulation for potential higher therapeutic effectiveness. This small molecule was obtained from a library created by combination and structural modification approach having triazole in them, with the functionalization of click chemistry to form a conjugate with polystyrene NPs [29]. According to the prior findings of this molecule, the small molecule RZA15 creates a hydrophilic outer layer on the surface of the polystyrene NPs that reduces the recruitment and attachment of the macrophages, which leads to a reduction in the recognition of particles by the macrophages.

Table 1. The pharmacokinetic parameters of P, PP, and PR after i.v. treatment in mice. The data is represented in the form of mean $\pm$ SD.

PK parameter	Unit	Mean (P) $\pm$ SD	Mean (PP) $\pm$ SD	Mean (PR) $\pm$ SD
$t_{1/2}$	min	20.78 $\pm$ 0.55	46.23 $\pm$ 0.22	28.48 $\pm$ 6.99
T_{max}	min	20 $\pm$ 0	15 $\pm$ 5	20 $\pm$ 0
C_{max}	Particles ml ⁻¹	4.2 $\times 10^3 \pm 0$	4.31 $\times 10^3 \pm 6.46 \times 10^2$	6.92 $\times 10^3 \pm 1.80 \times 10^3$
AUC 0– t	Particles ml ⁻¹ *min	1.9 $\times 10^5 \pm 2.1 \times 10^3$	1.98 $\times 10^5 \pm 1.2 \times 10^4$	2.64 $\times 10^5 \pm 5.35 \times 10^4$
AUC 0–inf_obs	Particles ml ⁻¹ *min	2.2 $\times 10^5 \pm 2.83 \times 10^2$	3.25 $\times 10^5 \pm 1.22 \times 10^4$	3.97 $\times 10^5 \pm 1.45 \times 10^5$
MRT 0–inf_obs	Min	37.05 $\pm$ 0.95	66.79 $\pm$ 2.74	49.29 $\pm$ 12.20
V_d _obs	ml ⁻¹ (particles ml ⁻¹) ⁻¹	1.33 $\times 10^7 \pm 3.72 \times 10^5$	2.05 $\times 10^7 \pm 8.69 \times 10^5$	1.08 $\times 10^7 \pm 1.43 \times 10^6$
CL_obs	ml ⁻¹ (particles ml ⁻¹) ⁻¹ min ⁻¹	4.45 $\times 10^5 \pm 5.62 \times 10^2$	3.07 $\times 10^5 \pm 1.15 \times 10^4$	2.90 $\times 10^5 \pm 1.06 \times 10^5$

For each group four mice were studied ($n = 4$). Here $t_{1/2}$, half-life; t_{max} , maximum time to reach peak plasma concentration; C_{max} , maximum concentration; AUC, area under the curve at time (0–infinity); MRT, mean residence time (0–infinity); V_d , apparent volume of distribution; CL, clearance.

Hence, modifying the surface of the nanoparticles using this novel class of small molecule can help in reducing macrophage uptake [11, 16].

Recent studies revealed that the pegylation of NPs enhances the retention time and blood flow, reducing non-specific reactions and avoiding fast clearance [30]. The pegylation of NPs is also reported with an increase in the surface hydrophilicity, which leads to a decrease in the phagocyte affinity [31]. As a result, it can avoid recognition by the RES and its capture by macrophages [32]. However, non-degradable PEG may aggregate and favor oxidation, limiting its long-term usage [33]. Hence, a new class of small molecules like RZA15 can have major advantages in reducing RES capture and improving particle circulation in the blood resulting in the potential therapeutic benefits.

It is crucial to highlight that here, PP were used as a structural positive control to the (PR) to keep the molecular weight of these particles consistent. RZA15 has PEG moieties in its structural backbone (scheme 1). It is evident that the control PEG-coated molecule (PP) did not significantly impact the NPs uptake and kinetics compared to the control P and PR group. Maisha *et al* demonstrated the importance of PEG attachment on the surface of the NPs for strengthening the stealth property. The MW and density of PEG can alter the conformation of PEG chains, leading to a change in the stealth property of PEG [34]. In this work, they used PEG 5000 and PEG 3400 (MW of 5 K and 3.4 K respectively). Varying PEG chain lengths with a higher molecular weight and increased hydrophilicity resulted in an effective stealth property. In another work, pegylated NPs (MW 2 K) having stealth properties were shown to retain the macrophage for efficient delivery of drugs [35]. Hence, we hypothesize that due to the lower MW (~ 400) of the PEG, we did not observe an effective stealth property of PEG when coated to the polystyrene NPs.

Reduced macrophage uptake has been reported by biodistribution studies *in vivo*, particularly in organs with higher macrophage concentrations, such as the liver and spleen [26]. PK parameters of RZA15 coated NPs (PR) further illustrate maximum concentration potency for effective and immediate therapeutic action. It also suggests a sustained elimination rate, which can help the formulation to stay inside the body for effective and prolonged action.

It is important to predict the mechanism of RZA15, preventing macrophage uptake, elevated particle circulation in the blood, and reduced biodistribution in the liver and spleen. Previous studies by Vegas *et al* demonstrated that RZA15 coating to the polymeric surface has significantly lowered macrophage recruitment near the polymeric capsules [11]. Intraperitoneal intravital imaging was carried out for 7 d after fluorescent microcapsule implantation in transgenic MAFIA mice (where macrophages produce green fluorescent protein (GFP)) to see whether lower macrophage recruitment was visible *in vivo*. The macrophage populations in small molecule (RZA15) functionalized microcapsules were significantly lower, and there was no evidence of macrophage aggregation [11]. This paper then served as instrumental in the development of a new generation of novel coating molecules that reduce inflammation, macrophage recognition, and attachment when coated to polymeric capsules [16]. These published works strongly support the mechanism of RZA15 coating in the reduction of macrophage recruitment and capture.

The major significance of this study illustrated that the NPs coated with RZA15 (PR) showed a reduction in non-specific organ uptake in the liver and spleens due to a decrease in macrophage recognition and capture. Later studies may include dose optimization for safety and efficacy in drug delivery. Future studies may evaluate the unexplored aspects of structural features of the newly developed small molecule

by varying the PEG changing length, increasing the hydrophilicity, and reducing the macrophage uptake. Hence, the potential of such material is not limited and can be used for various biomedical applications, including in treatments involving DDSs and implants.

5. Conclusion

The novel class of small molecule (RZA15) has shown effectiveness in reduced macrophage uptake of nanoparticles upon the molecular coating on the polystyrene NP. Additionally, initial *in vivo* studies were performed to demonstrate a reduction of particle distribution in the liver and spleen following RZA15 coating to polystyrene NPs resulting in the improvement of the blood circulation and slow-sustained release. Hence, in this proof-of-concept study, we have shown the utility of immunomodulation strategies in manipulating particle circulation in the body. Moreover, extensive therapeutic studies need to be performed to prove the utility of these molecular coatings, which is beyond the scope of this present study. We predict this novel small molecule coating will be used for future DDSs improving the NPs circulation, and therapeutic efficiency and reducing the non-specific distribution to other organs.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Authors contribution

Conceptualization: S M; experimentation by M R, D Y, S M and L L; writing-original draft preparation by M R, D Y, P, and S M. S M has corrected and supervised the work.

Conflict of interest

There are no conflicts to declare.

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